

1. Show that for any events $A_1, \dots, A_n$ we have

$$P(A_1 \cap \dots \cap A_n) = \prod_{i=1}^n P(A_i \mid A_{i-1} \cap \dots \cap A_1)$$

Solution. Inductively apply the definition of conditional probability

$$\begin{aligned} & P((A_1 \cap \dots \cap A_{n-1}) \cap A_n) \\ &= P(A_n \mid A_1 \cap \dots \cap A_{n-1}) P(A_1 \cap \dots \cap A_{n-1}) \\ &\dots \\ &= \prod_{i=1}^n P(A_i \mid A_{i-1} \cap \dots \cap A_1) \end{aligned}$$

□

2 (Feller V.8-11). Let the probability p_n that a family has exactly n children be αp^n when $n \geq 1$ and $p_0 = 1 - \alpha p(1 + p + p^2 + \dots)$. Suppose that all sex distributions of n children have the same probability.

- (a) Find the probability that a family has exactly k boys where $k \geq 1$
- (b) Given that a family includes at least one boy, what is the probability that there are two or more?

Proof. Let us do part (a) first. We can apply the law of total probability

$$\begin{aligned} P(k \text{ boys}) &= \sum_{n \geq k} P(k \text{ boys in } n \text{ children}) \\ &= \sum_{n \geq k} P(k \text{ boys} \mid n \text{ children}) P(n \text{ children}) \end{aligned}$$

We are given that $P(n \text{ children}) = p_n$. Since all sex distributions are equiprobable, and there are 2^n such distributions when there are n children, we can say that

$$P(k \text{ boys} \mid n \text{ children}) = 2^{-n} \binom{n}{k}.$$

Therefore

$$\sum_{n \geq k} \alpha \left(\frac{p}{2}\right)^n \binom{n}{k} = 2\alpha p^k / (2 - p)^{k-1}.$$

[Sidenote: In this course you're not expected to evaluate such sums, though its sort of a useful skill. If you are curious, you can try to do it by using

$$\frac{1}{1-x} = \sum_{n \geq 0} x^n$$

and differentiating k times.]

Let us do part (b). From the previous result we can compute $P(\text{boys} \geq i)$ for all i

$$\begin{aligned}
 P(\text{boys} \geq i) &= 2\alpha \sum_{k \geq i} \frac{p^k}{(2-p)^{k-1}} \\
 &= \frac{2\alpha p^i}{(2-p)^{i-1}} \sum_{k \geq 0} \left(\frac{p}{2-p}\right)^k \\
 &= \frac{2\alpha p^i}{(2-p)^{i-1}} \frac{1}{1 - \frac{p}{2-p}} \\
 &= \frac{\alpha p^i}{(2-p)^{i-2}(1-p)}
 \end{aligned}$$

Therefore

$$P(\text{boys} \geq 2 \mid \text{boys} \geq 1) = \frac{P(\text{boys} \geq 2)}{P(\text{boys} \geq 1)} = \frac{p}{2-p}$$

□

3 (Morin 2.7). Two people each proofread the same book. One person finds 100 errors, and the other finds 60. There are 20 errors common to both people. Assume that all errors are equally likely to be found (which is undoubtedly not true in practice), and also that the discovery of an error by one person is independent of the discovery of that error by the other person. Given these assumptions, roughly how many errors does the book have?

Solution. Suppose that n is the total number of errors. Letting A_i denote the event that the i th person finds an error, we estimate that $P(A_1) = 100/n$ and $P(A_2) = 60/n$. For these two events to be independent, we must have

$$\frac{20}{n} = P(A_1 \cap A_2) = P(A_1) P(A_2) = \frac{60 \cdot 100}{n^2}$$

which implies that $n = 300$.

□

References

- [1] William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1, 3rd ed., Wiley, 1968.
- [2] David J. Morin, *Probability: For the Enthusiastic Beginner*, CreateSpace, 2016.